

MOHAMMED AYAD

Full Stack Developer

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PROFILE SUMMARY

Full Stack Developer specializing in **MERN** stack, cloud-native architectures, and scalable distributed systems. Experienced in building real-time applications with **WebSocket/WebRTC** and deploying production-grade infrastructures on **AWS using Docker, Kubernetes (EKS), GitOps, Argo CD, and CI/CD pipelines**. Strong background in scalable backend engineering, DevOps automation, and high-availability system design.

SKILLS

- **Languages:** JavaScript, TypeScript, HTML, CSS,
- **Frontend:** React.js, Redux Toolkit, Tailwind CSS, Bootstrap
- **Backend:** Node.js, Express.js, REST APIs, WebSocket, WebRTC, Redis Pub/Sub
- **Database:** MongoDB, PostgreSQL, Firestore, Redis
- **Cloud & DevOps:** AWS (EKS, ECS, ECR, EC2, S3, CloudFront, ACM, ELB/ALB, Auto Scaling Groups, IAM, VPC), Kubernetes, Docker, Docker Compose, GitOps, Argo CD, Nginx Ingress Controller, CI/CD using GitHub Actions, Azure VM, Linux, PM2
- **Tools:** Git, GitHub, Postman, JWT, Axios, Prisma, Cloudinary, Figma, Pm2.

PROJECTS

Townify – Real-Time 2D Metaverse Platform [Live Code](#) [Gitops](#)

Developed a **real-time virtual collaboration platform** with room-based interactions and event-driven architecture.

- Engineered real-time user synchronization and presence system using **WebSocket**
- Built **interactive 2D environments** using Phaser.js with collision detection and tile-based navigation
- Integrated **WebRTC-based audio/video communication using SFU architecture** to support efficient multi-user streaming
- Designed backend using **Node.js, TypeScript, Prisma, and Redis Pub/Sub** for event-driven communication
- Structured project as a **monorepo using Turborepo** for scalable code organization

Deployment & Infrastructure:

- Deployed and managed applications on **Linux VM infrastructure with Nginx, SSL, and PM2**
- Built containerized cloud deployments using **Docker, AWS ECS, and Amazon ECR**
- Orchestrated production workloads on **AWS EKS with Kubernetes**, Nginx Ingress, ConfigMaps, HPA, and Secrets
- Implemented scalable infrastructure using **ALB, and Auto Scaling Groups (ASG)**

- Established **GitOps-based deployments using Argo CD and Git** as the single source of truth
- Automated **CI/CD pipelines using GitHub Actions** for build, testing, and production deployments

Tech Stack: MERN, TypeScript, WebSocket, WebRTC, Redis, Prisma, Redux Toolkit, Phaser.js, Tailwind CSS, OAuth, Turborepo, Docker, Kubernetes, AWS EKS, ECR, ECS, Argo CD, GitOps, GitHub Actions, Nginx Ingress, Redis Pub/Sub, Azure VM, PM2, Certbot, ALB, IAM

Learning Management System (LMS) Live Code

Built a production-grade LMS platform supporting Student, Instructor, and Admin workflows with integrated payments.

- Developed **secure authentication system** with OTP verification and JWT authorization
- Implemented **core features**: course enrollment, cart, wishlist, Razorpay payments
- Designed **analytics dashboards with revenue tracking**

Deployment & Infrastructure:

- Containerized application using **Docker for consistent deployments**
- Built **end-to-end CI/CD pipelines** using GitHub Actions and AWS CodePipeline
- Automated **image build, push to ECR, and deployment to ECS**
- Configured **ALB with Target Groups and health checks**
- Implemented **Auto Scaling Groups (ASG)** for scalability and fault tolerance
- Deployed frontend using **S3 + CloudFront for global delivery**

Tech Stack: MERN, TypeScript, Redux Toolkit, Tailwind CSS, Razorpay, AWS (EC2, S3, CloudFront, ACM, CodePipeline, CodeBuild, ECS, ECR, ASG), Docker, CI CD Pipeline (GitHub Actions, CodePipeline, CodeBuild), Cloudinary, Pm2

EXPERIENCE

MERN Stack developer Intern

Bridgeon Solution Aug 2025 - present

- Gained hands-on experience with **React, TypeScript, Redux Toolkit, Node.js, Express.js, MongoDB, PostgreSQL, Prisma, WebSocket, WebRTC**, and REST API development.
- Worked on building full-stack features, debugging issues, writing reusable components, and improving backend API performance.
- Collaborated with the team using Git/GitHub, participated in code reviews, and improved communication, teamwork, and problem-solving skills.
- Learned fundamentals of deploying and maintaining applications, schema design, authentication workflows, and modern development best practices.

EDUCATION

Bachelor of Computer Application

Yenepoya University 2022 - 2025